TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Catalin Picu	Email	picuc@rpi.edu
Institute Name	Rensselaer Polytechnic Institute			
Address	110 8th St., Troy, NY 12180, USA			

2. Technology Overview

Title of Proposed Technology	Manufacturing robust self-healing UV-cured thermoset coatings			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification 1)	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Chemicals		*Mechanics/Materials*	
	Medium Category	Small Category	Medium Category	Small Category
	Fine chemicals	*Coatings*	*Surface treatments*	*Coating/Method technologies*
Concept of Proposed Technology	Polymeric coatings are used in many applications for a variety of functionalities such as waterproofing, anticorrosion, UV protection, scratch resistance, modification of optical properties, etc. Applications include building envelopes, textiles and garments, microelectronics, furniture and wood products, and many others. Coatings are expected to be robust (i.e., to be stiff, strong, scratch resistant, adhere to the substrate) and to conform with the flexibility and function of the substrate (i.e., be hard and stiff when applied to a stiff substrate and be compliant when applied to a soft, deformable substrate). From a technological standpoint, coatings should be fast-drying and easy to apply (low viscosity before curing). UV-sensitive compositions fulfill these requirements. The proposed technology focuses on producing UV-sensitive coatings with embedded multifunctionality: high stiffness/strength, scratch resistance, self-healing, de-coating on demand. This will be achieved by creating (i) multiphase microstructures and by (ii) multi-layering in the thickness direction. The multiphase microstructure will develop during curing by phase separation			

(photo-polymerization-induced phase separation, photo-PIPS (1,2)). The two phases forming the coating will be thermosets with permanent bonds (acrylate-based) and thermosets with transient bonds (vitrimer), respectively. Adjusting the process parameters (UV flux, exposure time, intermittent irradiation) and the phase volume fractions we will adjust the combination of properties (mechanical, self-healing, reconfigurability). Multilayering may be achieved by 3D printing (digital light processing, DLP, or stereolithography, SLA) and will allow adjusting the composition and microstructure in the thickness direction, therefore further controlling coating functionality.

Vitrimers are permanent molecular networks at temperatures below the vitrification temperature, T_v, while above T_v bond exchange takes place, which allows network re-configuration and, at the macroscale, self-healing, shape memory, and the ability to flow plastically (3,4).

(i) The concept of the multiphase microstructure is shown in Fig. 1. The permanent bond thermoset and the transient bond vitrimer will phase separately during curing/polymerization to produce a complex microstructure. At ambient temperatures, the material is stiff and hard, with properties similar to acrylics. Above T_v, the vitrimeric domains undergo bond exchange, which allows them to flow and heal scratches and cracks. T_v is controlled by the material chemistry and may be adjusted to address the needs of specific applications. The objective of the proposed technical development is to achieve the desired combination of mechanical properties (see section 3/Target specifications) while maintaining self-healing and potentially enabling other functionalities required by the SME, such as color, transparency/opacity, etc.

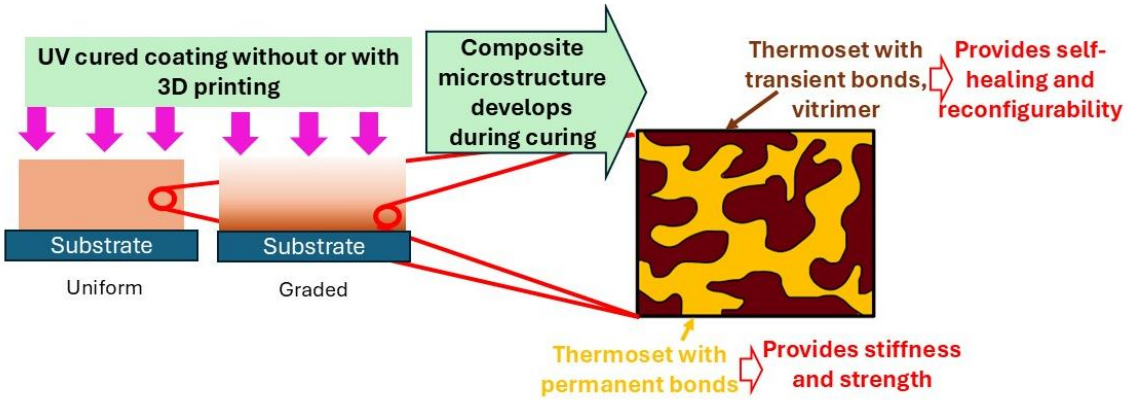

Figure 1. Proposed concept for the development of robust coatings with self-healing.

Similar methods may be used with soft materials (e.g., silicones), if the SME application requires soft or flexible substrates. In this case both permanent and transient networks are soft/elastomeric but have different functionalities.

(ii) The concept of multilayering is shown in Fig. 2. While nominally homogeneous coatings (Fig. 1) may be produced by either spinning or printing, multilayer coatings will be produced by printing (DLA or SLA). Multilayering creates functionality. For example, a top vitrimer-rich layer combined with a vitrimer-poor layer in contact with the substrate (or primer) allows for fast self-healing (Fig. 2a), while a vitrimer-rich layer in contact with the substrate allows for de-coating on demand (Fig. 2c). Combinations in which 2 types of vitrimers, with different T_v, are used at the top and at the interface are possible, as shown in Fig. 2b. A potential set of applications in the context of graphic protective coatings is shown schematically in Fig. 3.

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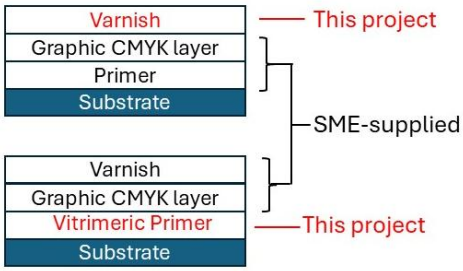
Figure 2. Potential configurations that provide different functionality. The vitrimer-poor layer provides mechanical properties, while the vitrimer-rich layers provide self-healing (in (a)) and de-coating on demand (in (c)).


Figure 3. Applications in the context of graphic protective coatings: textiles, shoes, furniture, building envelopes, etc.
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<td>Rationale for Joint R&D Project</td>
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⬭Rendering hard protective coatings self-healing. Self-healing and reconfigurability is a property of vitrimeric networks with transient bonds. However, <u>current UV curable vitrimers are soft, have low strength, and are not useful as protective coatings (5). The solution proposed here is to develop composite microstructures having a strong and stiff phase interpenetrated with a vitrimeric phase</u>, together providing stiffness and strength and self-healing properties (Fig. 1). Furthermore, it is desirable to <u>obtain the microstructure in one step, during curing</u>. This can be achieved by photopolymerization-induced phase separations (photo-PIPS). Our group has experience with both photo-PIPS (6-9) and vitrimers (10-14).
⬭Developing complex functionality in coatings and films. Adjusting the chemical composition and microstructure of coatings leads to multifunctionality, including mechanical properties, optical properties, self-healing and reconfigurability, humidity resistance, etc. Interaction with the SME is needed to define and prioritize the target properties.
⬭Collaboration between RPI and the Korean SME may evolve into a mutually beneficial synergetic partnership.
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<td>Key Words</td>
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1. Photo-polymerization
2. Thermoset
3. Self-healing
4. Protective coating
5. Mechanical properties
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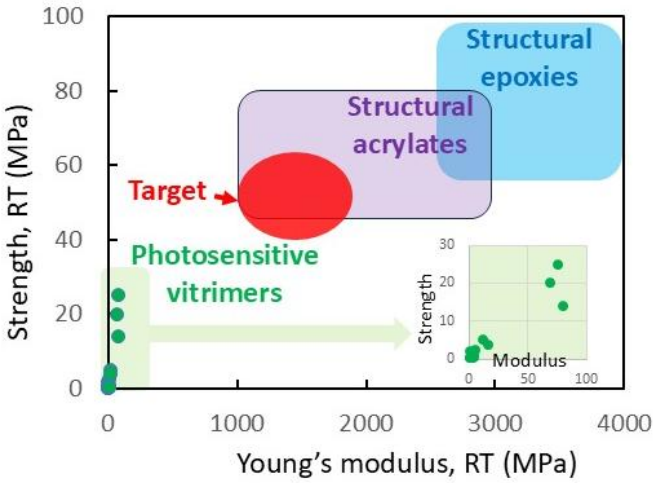
3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5)
R&D Objectives	⬭ Objective 1: Develop non-vitrimeric + vitrimeric compositions that phase separate effectively by photo-PIPS. ⬭ Objective 2: Identify process parameters and (vitrimeric & non-vitrimeric) phase fractions that allow control of multifunctionality. ⬭ Objective 3: Demonstrate target combinations of properties in non-printed (spun) coatings in a lab setting. ⬭ Objective 4: Demonstrate coatings with graded through thickness microstructure and properties produced by 3D printing. ⬭ Objective 5: Demonstrate these advances in the context of specific applications of SME.			
R&D Target Specifications	Photosensitive vitrimers have been developed over the past decade (3-5). However, they are typically soft, with stiffness below ~200 MPa and strength below ~30 MPa. Figure 4 shows the range of strength and moduli of commercial acrylates and epoxies (not self-healing or reconfigurable) and of current UV-sensitive vitrimers (light green). The present project aims to bridge the gap between these material classes creating composite microstructures with vitrimeric properties and with mechanical properties in the domain shown in red. Furthermore, the composite microstructure will develop during curing without the need of an additional manufacturing step.  Figure 4. Target mechanical properties of proposed materials (red) compared with those of current photosensitive vitrimers (light green) and of non–self-healing epoxies and acrylates. Self-healing will be provided by the vitrimer phase at temperatures comparable to and above T_v. T_v, may be adjusted based on the vitrimer chemistry. The target T_v will be defined in collaboration with the SME based on the target application.			
R&D Plan	The proposed work has 4 tasks: Task 1: Develop non-vitrimeric + vitrimeric compositions that phase separate effectively by photopolymerization induced phase separation. This task addresses Objective 1. Task 2: Establish the relation between target functionality, process parameters and microstructure. This task addresses Objectives 2 and 3.			

Task 3: Develop coatings with graded through thickness microstructure and properties via 3D printing and evaluate self-healing and de-coating behaviors. This task addresses Objective 4.

Task 4: Demonstrate the behavior of the developed coatings in the context of the application proposed by the SME. This task addresses Objective 5.

Preliminary technology development

The proposed technological developments are based on our preliminary work on polymerization-induced phase separation (6-9) and on vitrimers (10-14).

Photo-PIPS: We performed photo-PIPS in multiple acrylate-based systems, both in the context of UV curing of films and with 3D printing (SLA and DLP). The composition typically includes at least two types of acrylate monomers and a polymer which plays the role of phase separating agent. As the acrylate polymerizes upon exposure to UV, the polymer separates creating polymer-rich subdomains. The size of these domains may be controlled by varying process parameters such as light intensity, the exposure time or the schedule of intermittent irradiation (7). Figure 5 shows an example of a system composed of a tetrafunctional acrylate and a monofunctional methacrylate (6). A linear chain additive is added to promote phase separation. The microstructure of the material without additive (Fig. 5b) shows no features, while the material with additive has a two-phase microstructure. In Fig. 5c, the additive was removed to evidence the morphology of the thermoset subdomains. Interestingly, an additional set of subdomains, smaller than the wavelength used to obtain these images are present, which led to a separate peak in the dynamic mechanical analysis (DMA) spectrum shown in Fig. 5a. The 3 peaks in the red curve (with additive) correspond to the additive-rich domains (left peak), the main network composed primarily of acrylate (right peak), and nanoscale subdomains, which are rich in methacrylate, respectively.

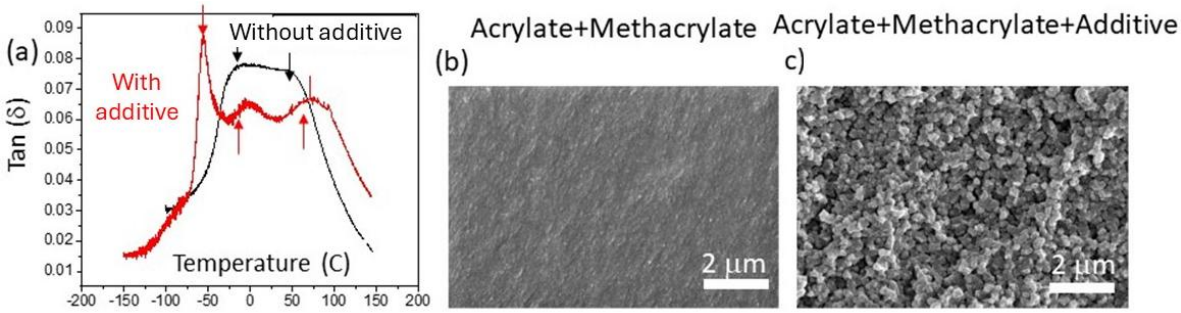


Figure 5. (a) DMA tan(δ) curves for materials with and without additive; SEMs of (b) acrylate+methacrylate and of (c) acrylate+methacrylate+additive after washing out the additive demonstrating phase separation. From (6).

Vitrimers: We worked with various types of vitrimer compositions including thermally cured and photosensitive, and explored their mechanical properties, ability to self-heal and reconfigure, and their shape memory (10-14). Figure 6a shows the results of tensile tests performed with a digycidyl ether+sebacid acid homogeneous vitrimer, which was used to produce samples that were cut, put back in contact and held above T_v for various durations and pressures (11). The cut was healed, and the strength of the repaired samples was dependent on the temperature and method used for healing. Figure 6b shows healing of a surface scratch in the same type of vitrimer (14).

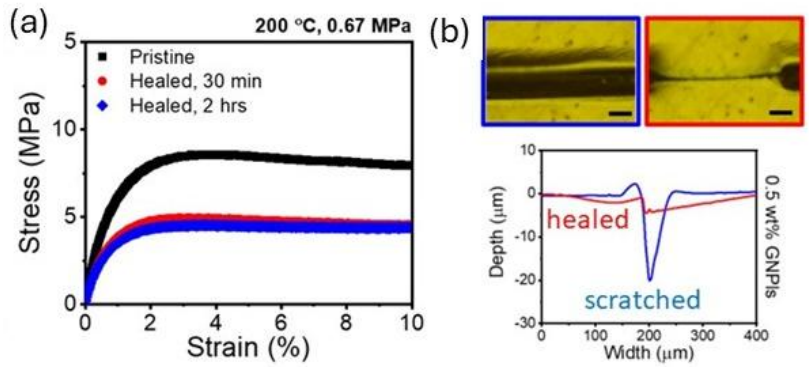


Figure 6. (a) Stress-strain curves before and after damage and healing, from (11), and (b) healing of a scratch in a digycidyl ether+sebacid acid vitrimer, from (14).

Proposed work

Task 1. Develop non-vitrimeric + vitrimeric compositions that phase separate effectively by photopolymerization induced phase separation

In this task, we seek to apply photo-PIPS to create composites with non-vitrimeric and vitrimeric subdomains. The goal is to phase separate the vitrimer and obtain separate subdomains that are rich in the non-vitrimeric component and are vitrimer-rich, respectively. The non-vitrimeric component will be acrylate-based. A range of diacrylates and tri-functional acrylates will be considered along with mono-acrylate monomers. The selection will be based on our prior work and the needs of the specific application proposed by the SME.

The vitrimer will be based on the trans-esterification reaction (3,10), which preserves the degree of crosslinking of the network. We will use multifunctional thiols along with acrylate monomers (5) and will control the concentration of moieties to adjust the rate of the transesterification reaction. Alternatives to the thiol-click chemistry are available and will be used as needed.

Task 2. Establish the relation between target functionality, process parameters and microstructure.

In addition to chemistry (see Task 1), the other parameters to be controlled are the volume fraction of the two phases and process parameters such as exposure time, flux and intermittent irradiation parameters (duration of pulses and time separation between pulses). Our preliminary work on this manufacturing technology provides guidance on the selection of process parameters (6-9).

For each combination of parameters, we will evaluate the microstructure, mechanical properties and the ability of the material to heal cuts and scratches, Fig. 6. The volume fraction of the two phases is a critical parameter which will affect multifunctionality. This will be varied in the entire 0%–100% range. The process parameters will determine the subdomain size and number density and is likely to be coupled with the phase volume fraction. We expect that obtaining sufficient strength and stiffness along with self-healing properties is achievable by controlling the composition and vitrimeric sub-domain size and volume fraction. These parameters will be varied in a broad range to develop an understanding of conditions in which the desired multifunctionality results.

Task 3. Develop coatings with graded through thickness microstructure.

3D printing (SLA or DLP) will be used to print coatings with composition, microstructure and combination of properties graded across the thickness. The three settings to be evaluated are shown in Fig. 2. The results of task 2 will guide the selection of the composition and UV exposure/processing parameters for each layer. For each of the 3 configurations of Fig. 2 we will evaluate mechanical behavior and self-healing properties.

De-coating primers (Fig. 2c) ease recycling by enabling coating removal leading to a clean substrate, which cuts disposal cost and facilitates recycling of substrates. A vitrimeric layer facilitates the removal of the coating. Furthermore, the vitrimer composition may be selected to facilitate solvolysis and decomposition in environmentally benign molecules. In situations such as Fig. 2b, where two vitrimeric layers with different functionalities are used, the T_v of the primer should be higher than the T_v of the varnish such to allow self-healing without de-coating.

Task 4. Demonstrate the behavior of both types of coatings in the context of the application proposed by the SME.

Tasks 1-3 will be guided by the specific problem proposed by the SME. The technology will be tested in the context of the respective application(s) and adjustments will be made to reach the technical targets specified by the SME.

Milestones and timeline:

Phase 1 (6 months): We demonstrated self-healing properties of vitrimeric compositions, Fig. 6, in agreement with the literature. In the first 6 months of the project, we will perform components of Task 1 and will address Objective 1. A subset of the previously explored compositions will be down-selected and will be used to develop phase separated vitrimeric and non-vitrimeric microstructures via photo-PIPS.

Phase 2:

12 months: Finalize Task 1 by focusing on specific compositions and reach Objective 1.

18 months: Begin exploring the relation between composition, process parameters, microstructure and multi-functionality, Task 2. Define the set of target properties (multi-functionality) together with the SME.

24 months: Finalize Task 2 which will lead to optimal compositions and process parameters that maximize the target properties, in the context of the SME application(s). Reach Objectives 2 and 3.

30 months: Implement Task 3 by developing 3D-printed coatings with through-thickness graded microstructure and properties; reach Objective 4.

36 months: Work with the SME on the target application(s), Task 4, and reach Objective 5.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	⬭Rensselaer Polytechnic Institute (RPI) will perform all research addressing Objectives 1 to 4. The specific tasks (Task 1 to 3) and steps to be taken are described under Milestones and timeline in section 3. ⬭RPI will consider a broad range of compositions and process parameters to maximize mechanical properties and self-healing. Target properties will be defined in collaboration with the SME. ⬭RPI will demonstrate the performance of 3D-printed coatings with nominally uniform and graded through thickness composition and structure.
Expected R&R of Korean SME Partner	⬭ The SME will interact with RPI in defining the application and target performance. ⬭ If the SME is interested in incorporating specific chemical components or functionalities beyond those discussed here (mechanical, self-healing), this information will be provided to RPI. If such components are not available commercially, the SME will provide these materials to RPI. ⬭ RPI and the SME will work together on Task 4 and share the responsibility of reaching Objective 5.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The outcomes of the proposed research are: 1) UV-curable thermosets (vitrimer+non-vitrimer mixtures) with a multiscale microstructure that conveys strength, stiffness, and self-healing properties beyond what can be obtained with each component separately. 2) The technology to produce such materials using a combination of photo-polymerization-induced phase separation and 3D printing (DLP or SLA). 3) Coatings structured in the through-thickness direction embedding multifunctionality such as strength and stiffness, scratch resistance and self-healing, and de-coating on demand. Other functionalities may be embedded as aligned best with the specific SME application. This technology applies to a variety of situations, including coatings for buildings (wall and window panels), furniture, shoes and other flexible commodities. The impact emerges from the added value to the application, i.e. extending the service life of the product on which the coating is applied. RPI will work with the SME on the implementation of this technology in the context of applications of interest for the Korean partner, while scaling, industrial impact and commercialization remain the responsibility of the SME.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization